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Thermal management in data storage device

Abstract

Thermal management of data storage devices in a server. In one example, a data storage device with a housing includes a first section, a second section, a third section, at least one first aperture that defines an airflow inlet, and at least one second aperture that defines an airflow outlet. A printed circuit board assembly ("PCBA") disposed in the housing includes passive components arranged in a first area corresponding to the second section and active components arranged a second area and a third area corresponding to the first section and the third section, respectively. The airflow inlet and the airflow outlet direct airflow through an airflow channel in the second section of the housing. The airflow channel has a first cross-sectional area between the PCBA and the second section of the housing that is larger than a second cross-sectional area between the PCBA and the first section of the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/491,768, filed on Mar. 23, 2023, the entire contents of which is incorporated herein by reference.

FIELD
(1) This application relates generally to server thermal management, and more specifically, the application relates to thermal management within data storage devices.
SUMMARY
(2) Information management systems ("IMS") or servers for enterprise generally consist of a

chassis system with multiple slots to accept data storage devices, such as solid-state drives ("SSDs"). Solid-state drives include a printed circuit board assembly ("PCBA") populated with active and passive components. As the number of components on the PCBAs increases, efficiency of heat dissipation within the SSD decreases. Additionally, active components often have differing heights and orientations, which causes turbulence in air flow through the SSD. Air flow produced by cooling fans in a server may not be able to flow through the SSD, thus forming an "inert" interior medium and further reducing efficiency of heat dissipation. These problems are typically addressed by replacing the thermal interface material of the PCBA with a thermal interface material having a higher conductivity grade, resulting in higher production costs. Additionally, applying thermal interface material onto active components will close free air gaps between the components and the housing, resulting in more air turbulence. Alternatively, the number of active components on a PCBA may be reduced, reducing performance of the SSD.
(3) Thus, the disclosure provides a data storage device including a housing and a printed circuit board assembly ("PCBA") disposed in the housing. The housing includes a first section, a second section, a third section, at least one first aperture that defines an airflow inlet, and at least one second aperture that defines an airflow outlet. The PCBA includes a plurality of passive components arranged in a first area corresponding to the second section and a plurality of active components arranged a second area and a third area corresponding to the first section and the third section, respectively. The airflow inlet and the airflow outlet direct airflow through an airflow channel in the second section of the housing. The airflow channel has a first cross-sectional area between the PCBA and the second section of the housing and a second cross-sectional area between the PCBA and the first section of the housing, the first cross-sectional area is larger than the second cross-sectional area.
(4) Another embodiment provides a data storage device including a housing and a PCBA disposed in the housing. The housing including a first section, a second section, a third section, a first aperture that defines an angled airflow inlet, and a second aperture that defines an angled airflow outlet. The first aperture is positioned between the first section and the second section, and the second aperture is positioned between the second section and the third section. The angled airflow inlet and the angled airflow outlet are configured to direct airflow through the second section of the housing. The PCBA includes a plurality of active components arranged in a first area corresponding to the second section and a plurality of passive components arranged in a second area and a third area corresponding to the first section and the third section, respectively. A second airflow inlet and a second airflow outlet are together configured to direct airflow through an airflow channel in the first section and third section of the housing. Each of the plurality of passive components are arranged adjacent to the airflow channel.
(5) Another embodiment provides a data storage device including a housing and a PCBA disposed in the housing. The housing includes a first section, a second section, a third section, at least one first aperture that defines an angled airflow inlet, and at least one second aperture that defines an angled airflow outlet. The PCBA includes a plurality of active components arranged in a first area corresponding to the second section and a plurality of passive components arranged a second area and a third area corresponding to the first section and the third section, respectively. The angled airflow inlet and the angled airflow outlet direct airflow through the second section of the housing. A first cross-sectional area between the PCBA and the first section is larger than a second cross-sectional area between the PCBA and the second section.
(6) Various aspects of the present disclosure provide for improvements in data storage devices, for example, improving thermal management in a server. The present disclosure can be embodied in various forms, including hardware or circuits and their placement on a PCBA. The foregoing summary is intended solely to give a general idea of various aspects of the present disclosure and does not limit the scope of the present disclosure in any way.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a block diagram of a server, in accordance with various aspects of the present disclosure.

(2) FIG. **2** illustrates a perspective view of a data storage device, in accordance with various aspects of the present disclosure.

(3) FIG. **3** illustrates a first example of a printed circuit board assembly included in a data storage device, in accordance with various aspects of the present disclosure.

(4) FIG. **4** illustrates a second example of a printed circuit board assembly included in a data storage device, in accordance with various aspects of the present disclosure.

(5) FIG. **5** illustrates a partial cutaway view of a data storage device, in accordance with various aspects of the present disclosure.

(6) FIG. **6**A illustrates a first cross-sectional view of a data storage device, in accordance with various aspects of the present disclosure.

(7) FIG. **6**B illustrates a second cross-sectional view of a data storage device, in accordance with various aspects of the present disclosure.

(8) FIG. **6**C illustrates a third cross-sectional view of a data storage device, in accordance with various aspects of the present disclosure.

(9) FIG. **7** illustrates a third example of a printed circuit board assembly included in a data storage device, in accordance with various aspects of the present disclosure.

DETAILED DESCRIPTION

(10) In the following description, numerous details are set forth, such as data storage device configurations, and the like, in order to provide an understanding of one or more aspects of the present disclosure. It will be readily apparent to one skilled in the art that these specific details are merely exemplary and not intended to limit the scope of this application. The following description is intended solely to give a general idea of various aspects of the present disclosure and does not limit the scope of the disclosure in any way. Furthermore, it will be apparent to those of skill in the art that, although the present disclosure refers to NAND flash, the concepts discussed herein are applicable to other types of solid-state memory, such as NOR, PCM (“Phase Change Memory”), ReRAM, etc.

(11) Terms of approximation, such as “generally,” “approximately,” or “substantially,” include values within five percent greater or less than the stated value. When used in the context of an angle or direction, such terms include within one degree greater or less than the stated angle or direction.

(12) FIG. **1** is a block diagram of a server system **100**, according to some embodiments. While the system **100** is described as a server system, it is understood that the system **100** may be an IMS system, or other system type which includes multiple data storage devices.

(13) The server system **100** includes system memory **102**, one or more power supplies **104**, an electronic processor **106**, a plurality of data storage devices **110**, one or more cooling fans **112** for facilitating air flow in the server system **100**, and a communication interface **114**. The communication interface **114** may include one or more communication devices, such as network interface devices. In one example, various information may be provided to, or requested from the one or more data storage devices **110** via the communication interface **114**. The request for retrieval of data from the data storage devices **110** and/or the storage of data to the data storage devices **110** may be processed by the one or more processors **106**. Generally, the server system **100** works as a general server system as required for a given application.

(14) The plurality of data storage devices **1110** are arranged in a chassis of the server system **100**. In one embodiment, the plurality of data storage devices **110** are solid-state drives (“SSD”), such as

non-volatile NAND SSDs. However, other SSD types are also contemplated. Additionally, in other examples, the plurality of data storage devices **110** may be other data storage devices, such as hard-disk drives ("HDD"). The cooling fans **112** are configured to direct air towards (e.g., push) the one or more data storage devices **110**. However, in other examples, the cooling fans **112** may be configured to direct air away (e.g., pull) from the data storage devices **110**, resulting in air flow being pulled across the data storage devices **110** in a direction away from the data storage devices **110**.

(15) The memory **102** may store thermal data **116** associated with the data storage devices **110**. For example, the memory **102** may periodically receive, from each of the plurality of data storage devices **110**, temperature information related to an internal temperature of the data storage device **200**. The thermal data **116** may include information associated with an amount of data operations performed by the plurality of data storage devices **110**. Alternatively, or in addition, the thermal data **116** may include information associated with an internal temperature of the plurality of data storage devices **110**. The electronic processor **106** may control the cooling fans **112** based on the thermal data **116**.

(16) FIG. **2** illustrates a perspective view of a data storage device **200**, included, for example, in the plurality of data storage devices **110** of FIG. **1**. The data storage device **200** includes a housing **204** having a first end surface **208** defining a first end of the housing **204** and a second end surface **212** opposite the first end surface **208** and defining a second end of the housing **204**. The first end surface **208** is arranged in a first plane, and the second end surface **212** is arranged in a second plane. In some examples, the first plane and the second plane may be substantially parallel to one another. The housing **204** includes an upper surface **216** arranged in a third plane traversing the first plane and the second plane. In some examples, the third plane may be substantially orthogonal to the first plane and the second plane. The upper surface **26** extends between the first end surface **208** and the second end surface **212**. The housing **204** also includes at least one side surface **220** arranged in a fourth plane substantially orthogonal to the upper surface **216** and either or both of the first end surface **208** and the second end surface **212**. The housing **204** may also include a lower surface opposite the upper surface **216**. The lower surface is arranged in a fifth plane traversing the first plane and the second plane. In some examples, the fifth plane is substantially parallel with the third plane.

(17) The first end surface **208** includes at least one first aperture that defines a first airflow inlet **224**. The first airflow inlet **224** is formed as an aperture in the first end surface **208** (e.g., a rectangular aperture, a square aperture, a circular aperture, or another shape). The first airflow inlet **224** receives airflow generated by the cooling fans **112** and directs the airflow through an interior of the housing **204**. In some instances, the first airflow inlet **224** includes more than one airflow inlets. For example, the first airflow inlet **224** may be formed as a plurality of apertures.

(18) The second end surface **212** includes at least one second aperture that defines a first airflow outlet **228**. The first airflow outlet **228** is formed as an aperture in the second end surface **212** (e.g., a rectangular aperture, a square aperture, a circular aperture, or another shape). The first airflow outlet **228** directs airflow out of the interior of the housing **204**. In some instances, the first airflow outlet **228** includes more than one airflow inlets. For example, the first airflow outlet **228** may be formed as a plurality of apertures.

(19) The housing **204** includes at least one third aperture that defines a plurality of second airflow inlets or angled inlets **232**. The angled inlets **232** include, for example, a first set of angled inlets **232***a* located on the upper surface **216** of the housing **204**, and a second set of angled inlets **232***b* located on the at least one side surface **220**. Like the first airflow inlet **224**, the angled inlets **232** are operable to direct airflow generated by the cooling fans **112** through the interior of the housing **204**. The plurality of angled inlets **232** are angled toward the interior of the housing **204** toward the second end of the housing **204**. The plurality of angled inlets **232** have an angle between approximately fifteen degrees and approximately seventy-five degrees (e.g., thirty degrees, forty-

five degrees, sixty degrees, etc.). The angle enables the plurality of angled inlets **232** to more efficiently direct air through the interior of the housing **204**, and therefore more efficiently provide heat dissipation in the data storage device **200**. For example, the angle of the plurality of angled inlets **232** enables air to enter the housing **204** at a higher speed and with less turbulence as compared with air entering a straight cut inlet at the upper surface **216** and/or the at least one side surface **220**. The housing **204** may include more angled inlets than just the first set of angled inlets **232***a*. Alternatively, or in addition to the first set of angled inlets **232***a*, additional angled inlets may be disposed on other surfaces, for example, the lower surface of the housing **204**.

(20) The housing **204** also includes at least one fourth aperture that defines a plurality of second airflow outlets, or angled outlets **236**. The angled outlets **236** include, for example, a first set of angled outlets **236***a* located on the upper surface **216** of the housing **204**, and a second set of angled outlets **236***b* located on the at least one side surface **220**. Like the first airflow outlet **228**, the angled outlets **236** are operable to direct airflow out of the interior of the housing **204**. The plurality of angled outlets **236** are angled toward the exterior of the housing **204** away from the first end of the housing **204**. Like the plurality of angled inlets **232**, the plurality of angled outlets **236** have an angle between approximately thirty degrees and approximately sixty degrees (e.g., thirty degrees, forty-five degrees, sixty degrees, etc.). The angle enables the plurality of angled outlets **236** to more efficiently direct air out of the interior of the housing **204**, and therefore more efficiently provide heat dissipation in the data storage device **200**. For example, the angle of the plurality of angled outlets **236** enables air to exit the housing **204** at a higher speed and with less turbulence as compared with air exiting a straight cut outlet at the upper surface **216** and/or the at least one side surface **220**. The housing **204** may include more angled outlets than just the first set of angled outlets **236***a*. Alternatively, or in addition to the first set of angled outlets **236***a*, additional angled outlets may be disposed on other surfaces, for example, the lower surface of the housing **204**.

(21) The housing **204** further includes a first section **238**, a second section **240**, and a third section **242**. The second section **240** is located between the first section **238** and the third section **242**. In the illustrated example, the second section **240** is located approximately between section lines A-A and C-C.

(22) The data storage device **200** includes a printed circuit board assembly ("PCBA") supported by the housing **204**. FIG. **3** illustrates a conventional example of such a PCBA, for example PCBA **300**. Conventional PCBAs such as PCBA **300** include a plurality of passive components **302** (e.g., resistors, capacitors, inductors, etc.) and a plurality of active components **304** (e.g., ball grid array ("BGA") packages). The plurality of active components **304** typically generate more heat than the plurality of passive components **302**. Additionally, the passive components **302** are typically smaller than the active components **304**. The larger size of the active components **304** relative to the passive components **302** increases air turbulence in the airflow path through the interior of the housing **204**, thus reducing the efficiency of heat dissipation in the data storage device **200**.

(23) Airflow turbulence is further increased by the location of the plurality of active components **304** in the conventional PCBA **300**. For example, the plurality of passive components **302** and the plurality of active components **304** are often randomly placed on the PCBA **300**. As air flows through the interior of the housing **204** and experiences turbulence from the larger active components **304**, the speed at which hot air, which has absorbed heat from the plurality of active components **304**, exits the interior of the housing **204** decreases, trapping hot air inside the data storage device **200**.

(24) Referring now to FIG. **4**, a first modified PCBA **400** is illustrated. The first modified PCBA **400** includes a plurality of passive components **402** and a plurality of active components **404**. The plurality of active components **404** are arranged in a central active component area **408** corresponding to the second section **240** of the housing **204**. The plurality of passive components **402** are arranged in an outer passive component area **412** corresponding to the first section **238** and the second section **240** of the housing **204**. For example, the passive component area **412** includes a

first passive component area **412***a* and a second passive component area **412***b*, with the active component area **408** disposed therebetween.

(25) The active component area **408** includes the majority of the active components included in the first modified PCBA **400**. For example, the active component area **408** may include at least approximately 80% of the active components included in the first modified PCBA **400**. Similarly, the passive component area **412** includes the majority of the passive components included in the first modified PCBA **400**. For example, the passive component area **412** may include at least approximately 80% of the passive components included in the first modified PCBA **400**. By providing the plurality of active components **404**, each of which may have a more similar height relative to one another than to the plurality of passive components **402**, in a single active component area **408**, air turbulence within the active component area **408** may be reduced.

(26) FIG. **5** illustrates a perspective view of a portion of the data storage device **200** including the first modified PCBA **400**. As illustrated in FIG. **5**, the plurality of angled inlets **232** are located on the first section **238** of the housing **204** approximately adjacent to a boundary between the first section **238** and the second section **240**. The plurality of angled outlets **236** are located on the third section **242** of the housing **204** approximately adjacent to a boundary between the second section **240** and the third section. The locations of the plurality of angled inlets **232** and the plurality of angled outlets **236** proximate to the boundaries of the second section **240** enable a shorter airflow path (e.g., airflow path **500**) through the active component area **408** than compared to the airflow path in a data storage device having only airflow inlets and outlets at either end of the data storage device **200** (e.g., the first airflow inlet **224** and the first airflow outlet **228** of FIG. **1**). A shorter airflow path through the active component area **408**, which includes the heat-generating plurality of active components **404**, allows for a higher rate of circulation of air from the exterior of the housing **204** through the interior of the housing **204**.

(27) FIG. **6**A illustrates a cross-sectional view of the first passive component area **412***a* taken along cross-sectional line A-A of FIG. **2**. FIG. **6**B illustrates a cross-sectional view of the active component area **408** taken along cross-sectional line B-B of FIG. **2**. FIG. **6**C illustrates a cross-sectional view of the second passive component area **412***b* taken along cross-sectional line C-C of FIG. **2**. As described above with respect to FIG. **4**, the first passive component area **412***a* and the second passive component area **412***b* include the smaller, passive components of the first modified PCBA **400**, while the active component area **408** includes larger, heat-generating components such as, for example, BGA packages. The active component area **408** is located in an area of the first modified PCBA **400** corresponding to the second section **240** of the housing **204**. The first passive component area **412***a* is located in an area of the first modified PCBA **400** corresponding to the first section **238** of the housing **204**. The second passive component area **412***b* is located in an area of the first modified PCBA **400** corresponding to the third section **242** of the housing **204**. Therefore, as illustrated in FIGS. **6**A-**6**C, a first cross-sectional area between the first modified PCBA **400** and the first section **238** is larger than a second cross-sectional area between the first modified PCBA **400** and second section **240**. Similarly, a third cross-sectional area between the first modified PCBA **400** and the third section **242** is larger than the second cross-sectional area.

(28) Air generated by the cooling fans **112** enters the first passive component area **412***a*, for example, via either or both of the first airflow inlet **224** and the angled inlets **232**, is heated as it passes through the active component area **408**, and flows into the second passive component area **412***b* before exiting through one or both of the first airflow outlet **228** and the angled outlets **236**. The air experiences a higher pressure as it travels through the first passive component area **412***a* and the second passive component area **412***b* relative to a lower pressure as it travels through the active component area **408** because the second cross-sectional area is smaller than the first cross-sectional area and the third cross-sectional area. The higher pressure in the active component area **408** causes the air to flow through the active component area **408** at a higher speed relative to that of the first passive component area **412***a* or the second passive component area **412***b*. For example,

the air may flow through the active component area **408** at a rate that is greater than or equal to approximately 295 linear feet per minute ("LFM"). The heated air particles are then pulled from the interior of the housing **204** through the angled outlet **236** located adjacent to the boundary of the second section **240** and the third section **242**. Enabling air to flow through the active component area **408**, which is populated with the heat-generating components of the first modified PCBA **400**, at a faster rate further improves heat transfer efficiency in the data storage device **200**.
(29) In some instances, each of the plurality of passive components **402** are arranged in the first passive component area **412***a* and the second passive component area **412***b* such that the plurality of passive components **402** defines an airflow channel, or airflow highway, through an approximate center of each of the first passive component area **412***a* and the second passive component area **412***b*, therefore further reducing air turbulence in the interior of the housing **204**. For example, the plurality of passive components may be arranged adjacent to an airflow channel extending along the approximate center of the first modified PCBA.
(30) Referring now to FIG. **7**, a second modified PCBA **700** is illustrated. The second modified PCBA **700** may be disposed in the housing **204** of the data storage device **200** alternative to the first modified PCBA **400**. The second modified PCBA **700** includes a plurality of active components arranged in a first active component area **708***a* and a second active component area **708***b*. The second modified PCBA **700** also includes a plurality of passive components arranged in a passive component area **712**.
(31) The first active component area **708***a* and the second active component area **708***b* include the majority of the active components included in the first modified PCBA **400**. For example, the first active component area **708***a* and the second active component area **708***b* may include at least approximately 80% of the active components included in the second modified PCBA **700**. Similarly, the passive component area **712** includes the majority of the passive components included in the second modified PCBA **700**. For example, the passive component area **712** may include at least approximately 80% of the passive components included in the second modified PCBA **700**.
(32) The passive component area **712** is located between the first active component area **708***a* and the second active component area **708***b* in a direction approximately perpendicular to an airflow direction through the interior of the housing **204**. For example, the passive component area **712** extends through the data storage device **200** in a longitudinal direction from the first end of the data storage device **200** to the second end of the data storage device **200**, and the passive component area **712** is located between the first active component area **708***a* and the second active component area **708***b* in a lateral direction.
(33) A cross-sectional area between the second modified PCBA **700** and the housing **204** in the first active component area **708***a* and the second active component area **708***b* is smaller than a cross-sectional area between the second modified PCBA **700** and the housing **204** in the passive component area **712**. The passive component area **412** therefore forms an airflow channel **716**, or highway, for directing air particles through the interior of the housing **204** while reducing the amount of turbulence in the airflow channel **716**.
(34) It is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provide would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.
(35) All terms used in the claims are intended to be given their broadest reasonable constructions and their ordinary meanings as understood by those knowledgeable in the technologies described

herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.
(36) The Abstract is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A data storage device comprising: a housing including a first section, a second section, a third section, at least one first aperture that defines an airflow inlet, and at least one second aperture that defines an airflow outlet, and a printed circuit board assembly (“PCBA”) disposed in the housing, the PCBA including a plurality of active components arranged in a first area corresponding to the second section and a plurality of passive components arranged a second area and a third area corresponding to the first section and the third section, respectively, wherein the airflow inlet and the airflow outlet direct airflow through an airflow channel in the second section of the housing, and wherein the airflow channel has a first cross-sectional area between the PCBA and the second section of the housing and a second cross-sectional area between the PCBA and the first section of the housing, the first cross-sectional area is larger than the second cross-sectional area.
2. The data storage device of claim 1, wherein the airflow channel has a third cross-sectional area between the PCBA and the third section of the housing, wherein the second cross-sectional area is larger than the third cross-sectional area.
3. The data storage device of claim 1, wherein the second section is located between the first section and the third section in a direction substantially perpendicular to an airflow direction, such that the airflow channel extends along an approximate center of the PCBA in the airflow direction.
4. The data storage device of claim 1, wherein the housing includes a first end and a second end opposite the first end, the airflow inlet is located at the first end, and the airflow outlet is located at the second end.
5. The data storage device of claim 1, wherein all passive components of the PCBA are strictly arranged in the second and third area on one or both of an upper surface of the PCBA and a lower surface of the PCBA.
6. A data storage device, comprising: a housing including a first section, a second section, a third section, a first aperture that defines an angled airflow inlet, and a second aperture that defines an angled airflow outlet, the first aperture positioned between the first section and the second section, and the second aperture positioned between the second section and the third section, the angled airflow inlet and the angled airflow outlet configured to direct airflow through the second section of the housing; a printed circuit board assembly (“PCBA”) disposed in the housing, the PCBA including a plurality of active components arranged in a first area corresponding to the second section and a plurality of passive components arranged in a second area and a third area corresponding to the first section and the third section, respectively; and a second airflow inlet and a second airflow outlet together configured to direct airflow through an airflow channel in the first section, the second section, and third section of the housing, wherein each of the plurality of passive components are arranged adjacent to the airflow channel.
7. The data storage device of claim 6, wherein the airflow channel extends along an approximate

center of the PCBA in an airflow direction.
8. The data storage device of claim 6, wherein the airflow channel has a first cross-sectional area between the PCBA and the first section, a second cross-sectional area between the PCBA and the second section, and a third cross-sectional area between the PCBA and the third section, wherein the first cross-sectional area is larger than the second cross-sectional area, and the third cross-sectional area is larger than the second cross-sectional area.
9. The data storage device of claim 6, wherein the second section is located between the first section and the third section in an airflow direction.
10. The data storage device of claim 6, wherein the plurality of active components are approximately 80% of all active components within the housing, and wherein the plurality of passive components is all passive components within the housing.
11. The data storage device of claim 6, wherein the angled airflow inlet is located approximately adjacent to a boundary between the first section and the second section, and the angled airflow outlet is located approximately adjacent to boundary between the second section and the third section.
12. The data storage device of claim 6, wherein an angle of the angled airflow inlet is between approximately 15 degrees and approximately 75 degrees, and an angle of the angled airflow outlet is between approximately 15 degrees and approximately 75 degrees.
13. The data storage device of claim 6, wherein the housing includes a first surface along a first plane and a second surface along a second plane that traverses the first plane, the angled airflow inlet includes a first angled airflow inlet located on the first surface and a second angled airflow inlet located on the second surface, and the angled airflow outlet includes a first airflow outlet located on the first surface and a second angled airflow outlet located on the second surface.
14. A data storage device comprising: a housing including a first section, a second section, a third section, at least one first aperture that defines an angled airflow inlet, and at least one second aperture that defines an angled airflow outlet, and a printed circuit board assembly (“PCBA”) disposed in the housing, the PCBA including a plurality of active components arranged in a first area corresponding to the second section and a plurality of passive components arranged a second area and a third area corresponding to the first section and the third section, respectively, wherein the angled airflow inlet and the angled airflow outlet direct airflow through the second section of the housing, and wherein a first cross-sectional area between the PCBA and the first section is larger than a second cross-sectional area between the PCBA and the second section.
15. The data storage device of claim 14, wherein a third cross-sectional area between the PCBA and the third section is larger than the second cross-sectional area.
16. The data storage device of claim 14, wherein the second section is located between the first section and the third section in an airflow direction.
17. The data storage device of claim 14, wherein the angled airflow inlet is located approximately adjacent to a boundary between the first section and the second section, and the angled airflow outlet is located approximately adjacent to boundary between the second section and the third section.
18. The data storage device of claim 14, wherein an angle of the angled airflow inlet is between approximately 15 degrees and approximately 75 degrees, and an angle of the angled airflow outlet is between approximately 15 degrees and approximately 75 degrees.
19. The data storage device of claim 14, wherein the housing includes a first surface along a first plane and a second surface along a second plane that traverses the first plane, the angled airflow inlet includes a first angled airflow inlet located on the first surface and a second angled airflow inlet located on the second surface, and the angled airflow outlet includes a first airflow outlet located on the first surface and a second angled airflow outlet located on the second surface.
20. The data storage device of claim 19, wherein the housing includes a third surface traversing the first surface and the second surface, the third surface defining a first end of the housing and a

fourth surface traversing the first surface and the second surface, the third surface defining a second end of the housing opposite the first end, the third surface includes at least one third aperture that defines an airflow inlet, and the fourth surface includes at least one fourth aperture that defines an airflow outlet.